

POOJA SUKHDEVE

AI Engineer | Data Analyst | Applied AI | Machine Learning

Remote-Ready | Open to Global Opportunities | [LinkedIn](#) | [Portfolio](#) | [Email:poojasukhdeve06@gmail.com](mailto:poojasukhdeve06@gmail.com)

Education

Boston University,

Master of Science in Computer Science (Data Analytics Concentration)

GPA: 3.59/4.00 | May 2026

Swami Ramanand Teerth Marathwada University

Bachelor in Computer Science Engineering

Jun 2016

Skills

- **Data Analysis, Visualization & cloud:** Power BI, Pandas, NumPy, Matplotlib, Seaborn, SHAP, A/B Testing, Data Wrangling, Data mining, Feature Engineering, Predictive Modelling, Business Intelligence, NLP, MySQL, Firebase, AWS, Apache Spark, OpenCV, OpenAI, GitHub.
- **Programming, Web & Software Development:** React, Node.js, Angular, RESTful APIs, Full-Stack Development, Responsive Design, Git), Agile Methodologies, SDLC, Python, SQL, JavaScript, TypeScript, HTML, CSS, PHP, Shell Scripting, Bash, MongoDB, Next.js, Supabase, and PostgreSQL.
- **Core & Interpersonal Skills:** Problem Solving, Team Collaboration, Analytical Thinking

Experience

Data Research Analyst | Boston University MET

Sept 2024 - Ongoing

- Designed 1,000+ lines of interactive Python exercises using data-driven methodologies, enhancing accessibility for visually impaired learners. Transformed complex programming concepts into tactile/visual learning tools including diagrams and flowcharts. Collaborated with faculty to research inclusive education, producing accessibility-compliant technical documentation in LaTeX.

Operations Analyst | Boston University IS&T

Sept 2024 - Oct 2025

- Automated data workflows using Excel (VLOOKUP, pivot tables, macros) across 1,000+ entry datasets, reducing reporting time by 40% and errors by 30%. Managed 350+ shipments and optimized inventory tracking systems, decreasing supply costs by 20% while achieving zero stockouts.

Software Developer Intern | ASVI Systems

May 2025 - Aug 2025

- Engineered 3 full-stack web applications with React.js, Next.js, and Node.js, serving 500+ daily users with JWT authentication and role-based access. Built 8+ RESTful APIs handling 10,000+ daily requests using Express.js, Flask, MongoDB, and PostgreSQL. Reduced page load times 35% through performance optimization. Automated deployment with Docker/GitHub Actions, cutting deployment time from 2 hours to 15 minutes.

Data Analyst | Tech Zenesis

Oct 2022 - Dec 2023

- Engineered scalable ETL pipeline using Python (Pandas, NumPy) and SQL, transforming data from 5+ sources for 50+ business users, reducing processing time by 50%. Built 8+ interactive Power BI dashboards with DAX calculations and real-time refresh, visualizing metrics across sales, marketing, and operations teams.

Software Developer | Interbeing Foundation

Apr 2020 - Jun 2022

- Developed 5+ full-stack applications using Python (Flask), Node.js (Express), and JavaScript, serving 10,000+ users with 50K+ monthly transactions at 99.9% uptime. Led Agile feature development, delivering 30+ features that improved user engagement by 40%. Optimized performance through code refactoring, reducing page load times by 45%.

Web Developer | Oceans Software Technologies

Apr 2018 - Mar 2020

- Designed user interfaces for 8+ banking web applications using HTML5, CSS3, and JavaScript for account management, loan applications, and transactions. Developed responsive layouts with CSS media queries, improving mobile experience by 30%. Built reusable HTML/CSS components ensuring consistent branding. Integrated frontend forms with MySQL and MongoDB databases using JavaScript validation.

Software Trainee – Android Development | SSIT Pvt. Ltd.,

Dec 2016 - Jun 2017

- Designed and developed Android and iOS applications using Java and Android SDK, implementing responsive UI/UX, RESTful API integration, SQLite database management, performance optimization, and ensuring code quality and version control using Git, JUnit, and Espresso.

Projects

Chatalogue – University Chatbot Assistant | [GitHub](#)

Designed and developed an AI-powered university assistant that enables students to interact with academic information using natural language. By combining GPT-4.1-mini, NLP, semantic search, and deterministic SQL generation, the system provides accurate, context-aware responses for course schedules, instructor lookup, and classroom navigation, showcasing practical applications of modern LLMs and AI-driven information retrieval.

SPY ETF Machine Learning Trading Strategy | [GitHub](#)

Explored whether historical market data can support data-driven investment decisions by developing an end-to-end machine learning trading system using SPY ETF data and technical indicators. Backtesting showed the Random Forest model achieved an **8.60%** cumulative return, **50.41%** positive trading days, and a Sharpe Ratio of **0.737**, while benchmark analysis revealed that a passive Buy & Hold strategy performed best, returning **24.99%** with a Sharpe Ratio of **1.897**.

AI-Powered Restaurant Analytics Dashboard | [GitHub](#) | [Live Demo](#)

Built an AI-powered restaurant analytics platform using Python, Flask, JavaScript, Chart.js, NLP, and the Hugging Face API to transform restaurant data into actionable insights. The application analyzes ratings, cuisines, and pricing trends from the Zomato India dataset through interactive dashboards, while a conversational AI assistant provides personalized restaurant recommendations and affordable dining suggestions based on natural language queries.

BRFSS Healthcare Predictive Analytics & Disease Risk Modeling | [GitHub](#)

Built an AI-powered healthcare analytics platform using the BRFSS dataset to predict disease risk and support data-driven health insurance decisions. Applied statistical analysis, feature engineering, and machine learning to evaluate **36 predictive models**, while SHAP-based explainability identified key risk factors. The best-performing Random Forest model achieved **84.3% accuracy**, demonstrating the practical value of interpretable AI in healthcare.

Publications & Achievements

- **Winner – Boston University Hackathon 2024:** Led development of a data science model generating actionable insights and measurable business impact.
- **Publication – Chapter 6: Implementing Augmented Reality into Immersive Virtual Learning Environments**, IGI Global, Pennsylvania, USA (2021).
- **Publication – Chapter 11: Handbook of Research on Driving Transformational Change in the Digital Built Environment**, IGI Global, Pennsylvania, USA (2021).